

## Topological Group.

Def. Let  $G$  be a Group under the binary operation  $\cdot: G \times G \rightarrow G$ .

A Topological Space  $(G, \mathcal{L})$  satisfying  $T_1$  condition, and

$$G \times G \rightarrow G: (x, y) \mapsto xy \text{ and } G \rightarrow G: x \mapsto x^{-1} \text{ are continuous}$$

is called the Topological Group.

Lemma. Let  $G$  be a group and  $T_1$ -space. Then

$G$  is Topological Group if and only if  $G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$  is continuous.

Proof. Suppose that  $G$  is Topological Group. Then

$\varphi(x, y) = xy$  and  $\psi(y) = y^{-1}$  are continuous, therefore

$(x, y) \mapsto xy^{-1} = \varphi(x, \psi(y))$  is continuous, being composition of continuous maps

Conversely, suppose that  $\mu: G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$  is continuous.

Then, for fixed  $x = 1$ ,  $\mu(1, y) = y^{-1}$  is continuous, so  $x \mapsto x^{-1}$  is continuous.

Now,  $y \mapsto y^{-1}$  is continuous, so

$(x, y) \mapsto x(y^{-1})^{-1} = xy$  is continuous, being composition of continuous.

~~QED~~

$GL_n(\mathbb{R})$

Consider  $GL_n(\mathbb{R})$  as the subset of  $\mathbb{R}^{n^2}$  equipped Euclidean Topology.

$GL_n(\mathbb{R})$ .

Consider  $GL_n(\mathbb{R})$  as the subset of  $\mathbb{R}^{n^2}$ , with Euclidean space.  
Define a Metric for  $M_{n \times n}(\mathbb{R})$  as:

$$d: M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}; (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu: M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R}); (A, B) \mapsto \left( \sum_{k=1}^n A_{ik} \cdot B_{ks} \right)_{i,j}$$

is continuous, because it is entry-wise continuous.

Since  $GL_n(\mathbb{R})$  is closed under the multiplication, it has also continuous operation.

And,  $GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R}); A \mapsto A^{-1}$

is also continuous, because  $A^{-1} = (\det A)^{-1} \cdot \text{adj } A$

$$\text{where } (\text{adj } A)_{i,j} = (-1)^{i+j} \cdot \det A(i|j).$$

is also multivariables polynomial, entrywise.

$\sqrt{SL}_n(\mathbb{R})$

Consider  $O_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^t = A^{-1}\}$ ,  $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$ .

Then,  $O_n$  and  $SO_n$  are compact, and  $SO_n$  is connected.

Proof.  $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^n A_{ik} \cdot A_{jk} = \delta_{ij}$ .

Define a map  $f_{ij}: M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}: A \mapsto \sum_{k=1}^n A_{ik} \cdot A_{jk}$ .

Then, it is continuous therefore

$O_n(\mathbb{R}) = \bigcap_{\substack{1 \leq i \leq n \\ i \neq j}} f_{ij}^{-1}[\{0\}] \cap \bigcap_{1 \leq i \leq n} f_{ii}^{-1}[\{1\}]$  is finite intersection of closed sets,

thus closed. Moreover,  $\dim O_n(\mathbb{R}) \leq n$ , so it is bounded. Therefore compact.

And,  $\det: O_n(\mathbb{R}) \rightarrow \mathbb{R}: A \mapsto \det A$  is continuous, so,

$SO_n(\mathbb{R}) = \det^{-1}[\{1\}]$  also closed, and bounded subset of bounded set.

$\Gamma_{\mathcal{O}_n}(\mathbb{R})$

$\Gamma_{SO_n}(\mathbb{B})$

$$\Gamma_n(\mathbb{C})$$